

Accessory Interface Hardware Installation Manual



OPERATING DOCUMENTATION

5317058-1EN

Rev 3

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IMPORTANT PRECAUTIONS

LANGUAGE

ПРЕДУПРЕЖДЕНИЕ (BG)	• ТОВА УПЪТВАНЕ ЗА РАБОТА Е НАЛИЧНО САМО НА АНГЛИЙСКИ ЕЗИК. • АКО ДОСТАВЧИКЪТ НА УСЛУГАТА НА КЛИЕНТА ИЗИСКА ЕЗИК, РАЗЛИЧЕН ОТ АНГЛИЙСКИ, ЗАДЪЛЖЕНИЕ НА КЛИЕНТА Е ДА ОСИГУРИ ПРЕВОД. • НЕ ИЗПОЛЗВАЙТЕ ОБОРУДВАНЕТО ПРЕДИ ДА СТЕ СЕ КОНСУЛТИРАЛИ И РАЗБРАЛИ УПЪТВАНЕТО ЗА РАБОТА. • НЕСПАЗВАНЕТО НА ТОВА ПРЕДУПРЕЖДЕНИЕ МОЖЕ ДА ДОВЕДЕ ДО НАРАНЯВАНЕ НА ДОСТАВЧИКА НА УСЛУГАТА, ОПЕРАТОРА ИЛИ ПАЦИЕНТ В РЕЗУЛТАТ НА ТОКОВ УДАР ИЛИ МЕХАНИЧНА ИЛИ ДРУГА ОПАСНОСТ.
警告 (ZH-CN)	• 本维修手册仅提供英文版本。 • 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。 • 未详细阅读和完全理解本维修手册之前，不得进行维修。 • 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。
VÝSTRAHA (CS)	• Tento provozní návod existuje pouze v anglickém jazyce. • V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. • Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. • V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
ADVARSEL (DA)	• Denne servicemanual findes kun på engelsk. • Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse. • Forsøg ikke at servicere udstyret medmindre denne servicemanual har været konsulteret og er forstået. • Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk, mekanisk eller anden fare for teknikeren, operatøren eller patienten.
WAARSCHUWING (NL)	• Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. • Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. • Probeer de apparatuur niet te onderhouden voordat deze onderhoudshandleiding werd geraadpleegd en begrepen is. • Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

WARNING (EN)	• This Service Manual is available in English only. • If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services. • Do not attempt to service the equipment unless this service manual has been consulted and is understood. • Failure to heed this warning may result in injury to the service provider, operator, or patient, from electric shock or from mechanical or other hazards.
HOIATUS (ET)	• Käesolev teenindusjuhend on saadaval ainult inglise keeles. • Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. • Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. • Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
VAROITUS (FI)	• Tämä huolto-ohje on saatavilla vain englanniksi. • Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. • Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. • Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
ATTENTION (FR)	• Ce manuel de service n'est disponible qu'en anglais. • Si le technicien du client a besoin de ce manuel dans une autre langue que l'anglais, c'est au client qu'il incombe de le faire traduire. • Ne pas tenter d'intervenir sur les équipements tant que le manuel service n'a pas été consulté et compris • Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.
WARNUNG (DE)	• Diese Serviceanleitung existiert nur in Englischer Sprache. • Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es aufgabe des Kunden für eine Entsprechende Übersetzung zu sorgen. • Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben. • Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, Mechanische oder Sonstige gefahren kommen.
ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)	• Το παρόν εγχειρίδιο σέρβις διατίθεται στα αγγλικά μόνο. • Εάν το άτομο παροχής σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει υπηρεσίες μετάφρασης. • Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό εκτός εάν έχετε συμβουλευτεί και έχετε κατανοήσει το παρόν εγχειρίδιο σέρβις. • Εάν δε λάβετε υπόψη την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στο άτομο παροχής σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

FIGYELMEZTETÉS (HU)	• Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. • Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése. • Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. • Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.
AÐVÖRUN (IS)	• Þessi þjónustuhandbók er eingöngu fánleg á ensku. • Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu. • Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin. • Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.
AVVERTENZA (IT)	• Il presente manuale di manutenzione è disponibile soltanto in inglese. • Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. • Si proceda alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto • Il non rispetto della presente avvertenza potrebbe far compiere operazioni da cui derivino lesioni all'addetto, alla manutenzione, all'utilizzatore ed al paziente per folgorazione elettrica, per urti meccanici od altri rischi.
警告 (JA)	• このサービスマニュアルには英語版しかありません。 • サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。 • このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。 • この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。
경고 (KO)	• 본 서비스 지침서는 영어로만 이용하실 수 있습니다. • 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다. • 본 서비스 지침서를 참고했고 이해하지 않는 한은 해당 장비를 수리하려고 시도하지 마십시오. • 이 경고에 유의하지 않으면 전기 쇼크, 기계상의 혹은 다른 위험으로부터 서비스 제공자, 운영자 혹은 환자에게 위해를 가할 수 있습니다.
BRDINJUMS (LV)	• Ģī apkalpes rokasgrāmata ir pieejama tikai angļu valodā. • Ja klienta apkalpes sniedzējam nepieciešama informācija citā valodā, nevis angļu, klienta pienākums ir nodrošināt tulkošanu. • Neveiciet aprīkojuma apkalpi bez apkalpes rokasgrāmatas izlasīšanas un saprašanas. • Ģī brīdinājuma neievērošana var radīt elektriskās strāvas trieciena, mehānisku vai citu risku izraisītu traumā apkalpes sniedzējam, operatoram vai pacientam.

ÁSPĖJIMAS (LT)	• Dis eksploataavimo vadovas yra prieinamas tik anglų kalba. • Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, numatyti vertimo paslaugas yra kliento atsakomybė. • Nemėginkite atlikti árangos techninės priežiūros, nebent atsipvelgęte á áá eksploataavimo vadovà ir já supratote. • Jei neatkreipsite dėmesio á áá perspėjimą, galimi subalojimai dėl elektros šoko, mechaninių ar kitų pavojų paslaugų tiekėjui, operatoriui ar pacientui.
ADVARSEL (NO)	• Denne servicehåndboken finnes bare på engelsk. • Hvis kundens serviceleverandør trenger et annet språk, er det kundens ansvar å sørge for oversettelse. • Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. • Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
OSTRZEŻENIE (PL)	• Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. • Jeśli dostawca usług klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. • Nie próbować serwisować wyposażenia bez zapoznania się i zrozumienia niniejszego podręcznika serwisowego. • Niezastosowanie się do tego ostrzeżenia może spowodować urazy dostawcy usług, operatora lub pacjenta w wyniku porażenia elektrycznego, zagrożenia mechanicznego bądź innego.
ATENÇÃO (PT)	• Este manual de assistência técnica só se encontra disponível em inglês. • Se qualquer outro serviço de assistência técnica solicitar estes manuais noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. • Não tente consertar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • O não cumprimento deste aviso pode pôr em perigo a segurança do técnico, do operador ou do paciente devido a choques elétricos, mecânicos ou outros.
ATENȚIE (RO)	• Acest manual de service este disponibil numai în limba engleză. • Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. • Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. • Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

ОСТОРОЖНО! (RU)	• Данное руководство по обслуживанию предлагается только на английском языке. • Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. • Перед обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. • Несоблюдение требований данного предупреждения может привести к тому, что специалист по обслуживанию, оператор или пациент получат удар электрическим током, механическую травму или другое повреждение.
UPOZORNENIE (SK)	• Tento návod na obsluhu je k dispozícii len v angličtine. • Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. • Nepokúšajte sa o obsluhu zariadenia skôr, ako si neprečítate návod na obsluhu a neporozumiete mu. • Zanedbanie tohto upozornenia môže vyústiť do zranenia poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, do mechanického alebo iného nebezpečenstva.
ATENCION (ES)	• Este manual de servicio sólo existe en inglés. • Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. • No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. • La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
VARNING (SV)	• Den här servicehandboken finns bara tillgänglig på engelska. • Om en kunds servicetekniker har behov av ett annat språk än engelska ansvarar kunden för att tillhandahålla översättningstjänster. • Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. • Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.
DIKKAT (TR)	• Bu servis kılavuzunun sadece ingilizcesi mevcuttur. • Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer. • Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz. • Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "damage in shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on <http://us44hdd21/sctq/InstallFulfill/InstalFulfillment.htm>
- Contact your local service coordinator for more information on this process.

Rev. Jan. 5, 2005

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All licensed electrical contractors shall perform all electrical de-installation work required to remove electrical feeds from the Power Distribution Unit and the A1 disconnect. The products involved are highly sophisticated, and special engineering competence is required. All de-installation electrical work must comply with the requirements of the applicable electrical codes. GE Healthcare (GEH) shall only utilize trained personnel (i.e., General Electric field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform the de-installation work.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the fundamentals of radiation, completed a course in Radiation Protection, and take adequate steps to protect themselves against injury. Various protective materials and devices are available. It is urged that such materials or devices be used.

The equipment is sold with the understanding that the General Electric Company, GE Healthcare, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GE Healthcare iTrak Process to report all omissions, errors, and defects in this publication.

Revision History

Revision	Date	Reason for change
3	10/27/08	Correct an error in writing.

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List of Effected Pages

PAGES	REVISION	PAGES	REVISION	PAGES	REVISION
1 through 38	3				

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Preface

Publication Conventions

Purpose: Standardize conventions for representing information in a uniform way of communicating information to a reader in a consistent manner. Conventions are used so that the reader can easily recognize the actions or decisions that must be made. There are a number of character and paragraph styles used in this publication to accomplish this task. Please become familiar with them before proceeding forward.

Section 1.0

Safety & Hazard Information

1.1 Text and Character Representation

Within this publication, different paragraph and character styles have been used to indicate potential hazards. Paragraph prefixes, such as hazard, caution, danger and warning, are used to identify important safety information. Text (Hazard) styles are applied to the paragraph contents that is applicable to each specific safety statement. Words describe the type of potential hazard that may be encountered and are placed immediately before the paragraph it modifies. Safety information will normally include:

- Type of potential Hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

EXAMPLES OF HAZARD STATEMENTS USED

It's important that you read and understand hazard statements, and not just ignore them.

DANGER SEVERE PERSONAL INJURY

DANGER IS USED WHEN A HAZARD EXISTS WHICH WILL CAUSE SEVERE PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- **ELECTROCUTION**
- **CRUSHING**
- **RADIATION**

WARNING SERIOUS PERSONAL INJURY

WARNING IS USED WHEN A HAZARD EXISTS WHICH COULD OR CAN CAUSE SERIOUS PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- **POTENTIAL FOR SHOCK**
- **EXPOSED WIRES**
- **FAILURE TO TAG AND LOCKOUT SYSTEM POWER COULD ALLOW FOR UN-COMMANDED MOTION.**

CAUTION Minor Personal Injury

Caution is used when a hazard exists which can or could cause minor injury to self or others if instructions are ignored. They include for example:

- **Loss of critical patient data**
- **Crush or pinch points**
- **Sharp objects**

NOTICE	Notice is used when a hazard is present that can cause property damage but has <u>absolutely no personal injury risk</u> . They can include:
Property	
Damage	• Disk drive will be destroyed • Internal mechanical damage, such as to the x-ray tube • Coasting the rotor through resonance.
Only	

1.2 Graphical Representation

Important information will always be preceded by the exclamation point  contained within a triangle, as seen throughout the Preface chapter. In addition to text, several different graphical icons (symbols) may be used to make you aware of specific types of hazards that could possibly cause harm.

ELECTRICAL



LASER



LASER LIGHT

HAZARDS

MECHANICAL



HEAT



RADIATION



PINCH



AVOID STATIC ELECTRICITY



PROCEDURAL

TAG AND LOCK OUT



WEAR EYE PROTECTION



EYE PROTECTION

Section 2.0
Publication Conventions

2.1 General Paragraph and Character Styles

Prefixes are used to highlight important non-safety related information. Paragraph prefixes (such as purpose, example, comment or note) are used to identify important but non-safety related information. Text styles are also applied to text within each paragraph modified by the specific prefix.

EXAMPLES OF PREFIXES USED FOR GENERAL INFORMATION

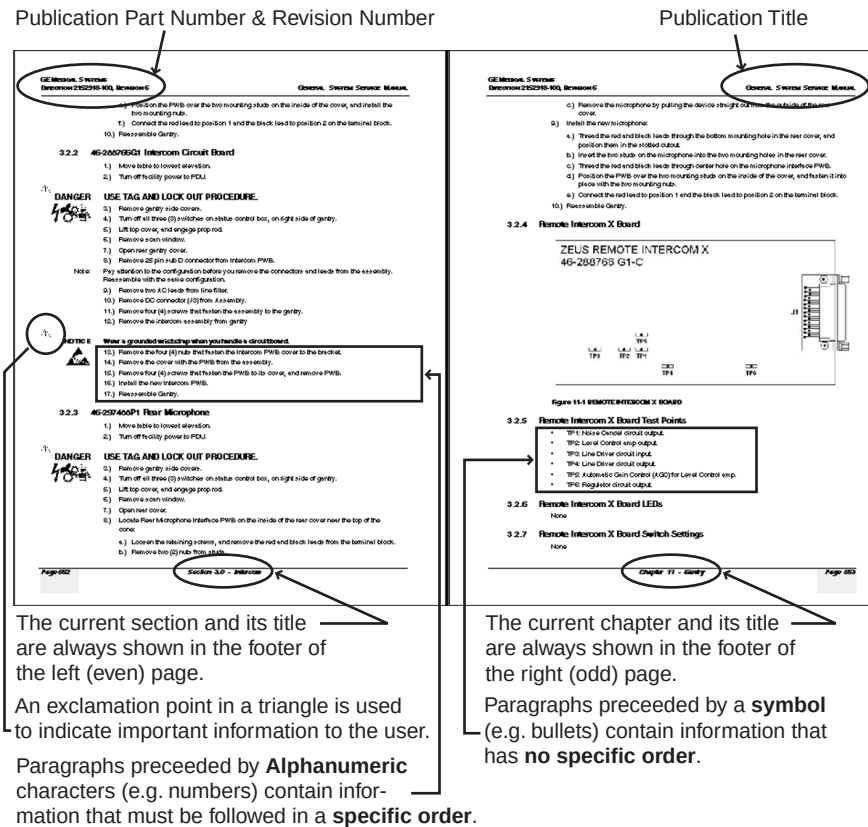
Purpose: Introduces and provides meaning as to the information contained within the chapter, section or subsection, Such as used at the beginning this chapter for example.

Note: Conveys information that should be considered important to the reader.

Example: Used to make the reader aware that the paragraph(s) that follow are examples of information possibly stated previously.

Comment: Represents “additional” information that may or may not be relevant to your situation.

2.2 Page Layout



Headers and footers in this publication are designed to allow you to quickly identify your location. The document's part number and revision number appears in every header on every page. Odd numbered page footers indicate the current chapter, its title and current page number. Even page footers show the current section and its title, as well current page number.

2.3 Computer Screen Output/Input Text Character Styles

Within this publication, mono-spaced character styles (fonts) are used to indicate computer text that's either screen input and output. Mono spaced fonts, such as courier, are used to indicate text direction. When you type at your keyboard, you are generating computer input. Occasionally you will see the math operators “greater than” and “less than” symbols used to indicate the start and finish of variable output. When reading text generated by the computer, you are reading it as computer generated output. In addition to direction, characters are italicized (e.g. *italics*) to indicate information specific to your system or site.

Example:
Fixed Output

This paragraph's fonts represents computer generated screen “fixed” output. It's output is fixed from the sense that it does not vary from application to application. It's the most commonly used style used to indicate filenames, paths and text that does not change from system to system. The character style used is a fixed width such as courier.

Example: Variable Output	<i>This paragraph's font represents computer screen output that is "variable". Its used to represent output that varies from application to application or system to system. Variable output is sometimes found placed between greater than and lesser than operators for clarification. For example: <variable_ouput> or <3.45.120.3>. In both cases, the < and > operators are not part of the actual input.</i>
Example: Fixed Input	This paragraph's font represents fixed input. It's computer input that is typed-in via the keyboard. Typed input that does not vary from application to application or system to system. Fixed text the user is required to supply as input. For example: cd /usr/3p
Example: Variable Input	<i>This paragraph's font represents computer input that can vary from application to application or system to system. With variable text, the user is required to supply system dependent input or information. Variable input sometimes is placed between greater-than and less-than operators. For example: <variable_input>. In these cases, the (<>) operators would be dropped prior to input. For example: ypcat hosts grep <3.45.120.3> would be typed into the computer as ypcat hosts grep 3.45.120.3 without the greater-than and less-than operators.</i>

2.4 Buttons, Switches and Keyboard Inputs (Hard & Soft Keys)

Different character styles are used to indicate actions requiring the reader to press either a hard or soft button, switch or key. Physical hardware, such as buttons and switches, are called hard keys because they are hard wired or mechanical in nature. A keyboard or on/off switch would be a hard key. Software or computer generated buttons are called soft keys because they are software generated. Software driven menu buttons are an example of such keys. Soft and hard keys are represented differently in this publication.

Example: Hard Keys	A power switch <u>ON/OFF</u> or a keyboard key like <u>ENTER</u> is indicated by applying a character style that uses both over and under-lined bold text that is bold. This is a hard key.
Example: Soft Keys	Whereas the computer <u>MENU</u> button that you would click with your mouse or touch with your hand uses over and under-lined regular text. This is a soft key.

END OF SECTION

Chapter 1

Accessory Interface Hardware Installation

Section 1.0

Introduction

1.1 Overview

The Accessory Interface Hardware option provides interface for the following options.

- Cardiac Gating
- Advantage 4D Respiratory Gating
- Smart Step
- Xstream Injector

The Accessory Interface Hardware option is shipped ready for immediate installation. The following installation procedure describes how to install that hardware on the Gantry.

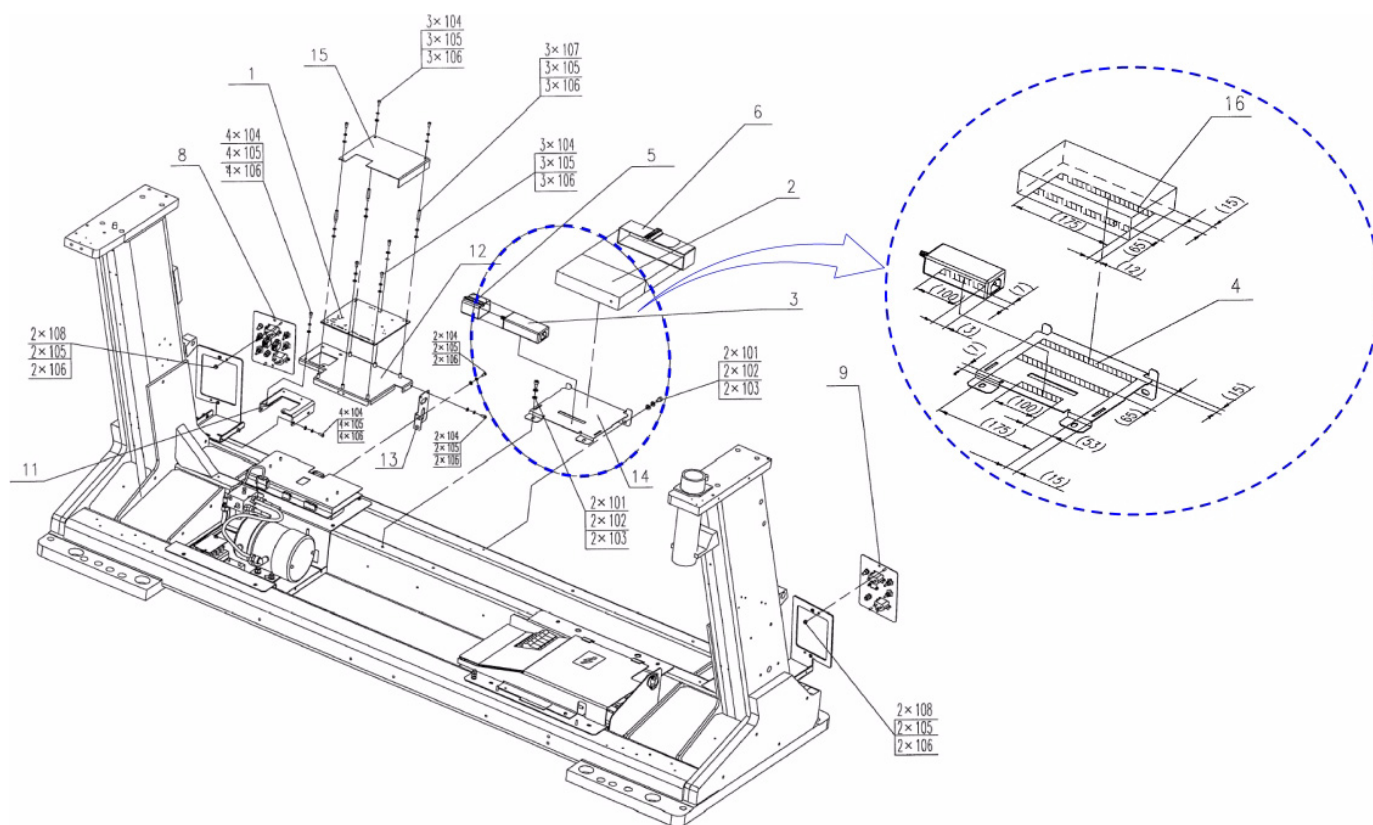
Refer to [Table 1-1](#) for a listing of the parts in the kit.

1.2 Prerequisites

- BrightSpeed and BrightSpeed Select Gantry with New Power Pan (5272458)

Section 2.0

Material List



Mark	Part Number	Quantity	Description
1	5136002	1.0	Integrated Peripheral Controller PWA
2	5115334	1.0	Switch, Ethernet 8-port GBe
3	5136650	1.0	ups-120vac to 12vdc,
4	46-136246P18	2.0	1~ WIDE, WHT, ADH BACK, VELCRO HOOK 75 FT PER RL
5	2315482	1.0	VELCRO TAPE LENGTH IS 320
6	2315482-3	1.0	VELCRO TAPE LENGTH IS 830
7	5316429	1.0	BSD V3 Cable Harness Assembly
8	5306106	1.0	Accessory Panel LH ASM, CT HD
9	5306107	1.0	Accessory Panel RH ASM, CT HD
10	5306201	1.0	Cable, Cardiac Pulmonary, CT HD
11	5309358	1.0	BSD V3 IPC MNT brkt cushion
12	5309359	1.0	BSD V3 IPC MNT brkt
13	5309361	1.0	BSD V3 connector MNT brkt
14	5309641	1.0	BSD V3 LAN Switch MNT BRKT
15	5304664	1.0	Cover IPC Board, Kitty Hawk
16	46-136246P16	2.0	1~ WIDE, WHT, ADH BACK, VELCRO LOOP 75 FT PER RL

Table 1-1 5317931 Material List

Mark	Part Number	Quantity	Description
17	46-208758P3	20.0	7.31~ X .184~ SELF-LOCKING CABLE TIE 1/16~ TO 1-3/4~ WIRE BUNDLE RANGE. 50LB TENSILE STRENGTH, NYLON MATR~L.
101	46-328417P12	4.0	SCREW HEXAGON SOCKET 6 MM 12 MM ZINC PLATED, STEEL
102	46-328432P3	4.0	WASHER LOCK - SPRING 6.1 MM 11.8 MM ZINC PLATED, SPRING STEEL
103	46-328431P2	4.0	WASHER PLAIN - LARGE 6.4 MM 18 MM ZINC PLATED, STEEL
104	46-328417P3	18.0	SCREW HEXAGON SOCKET 4 MM 10 MM ZINC PLATED, STEEL
105	46-328432P2	25.0	WASHER LOCK - SPRING 4.1 MM 7.6 MM ZINC PLATED, SPRING STEEL
106	46-328430P2	25.0	WASHER PLAIN - NORMAL 4.3 MM 9 MM ZINC PLATED, STEEL
107	U0099AG	3.0	STANDOFF/SPACER MALE/FEMALE 4 MM 15 FF15 STUD
108	46-328425P1	4.0	NUT HEXAGON 4 MM
	5317058-1EN	1	Instruction

Table 1-1 5317931 Material List

Section 3.0

Installation Procedure

DANGER RISK OF ELECTRIC SHOCK. USE PROPER LOCKOUT/TAGOUT PROCEDURES.

3.1 Base Cover Removal

- 1.) Tilt the Gantry backward approximately 20° to be able to access the Gantry rear base components.
- 2.) Remove the two gantry side covers. Refer to your System Service Manual for any needed instruction.
- 3.) Turn **OFF** all 3 switches (Axial Drive, HVDC, 120 VAC) on the service switch panel.
- 4.) Shutdown system software and turn off the console power switch under the console table top.
- 5.) Remove all system power at the main disconnect (A1) panel. Perform proper Lockout/Tagout power control procedures.
- 6.) Remove both left and right Gantry side base covers (items 6 & 7, respectively, in [Figure 1-1](#)) and then both rear base covers (items 8 & 9, in [Figure 1-1](#)).

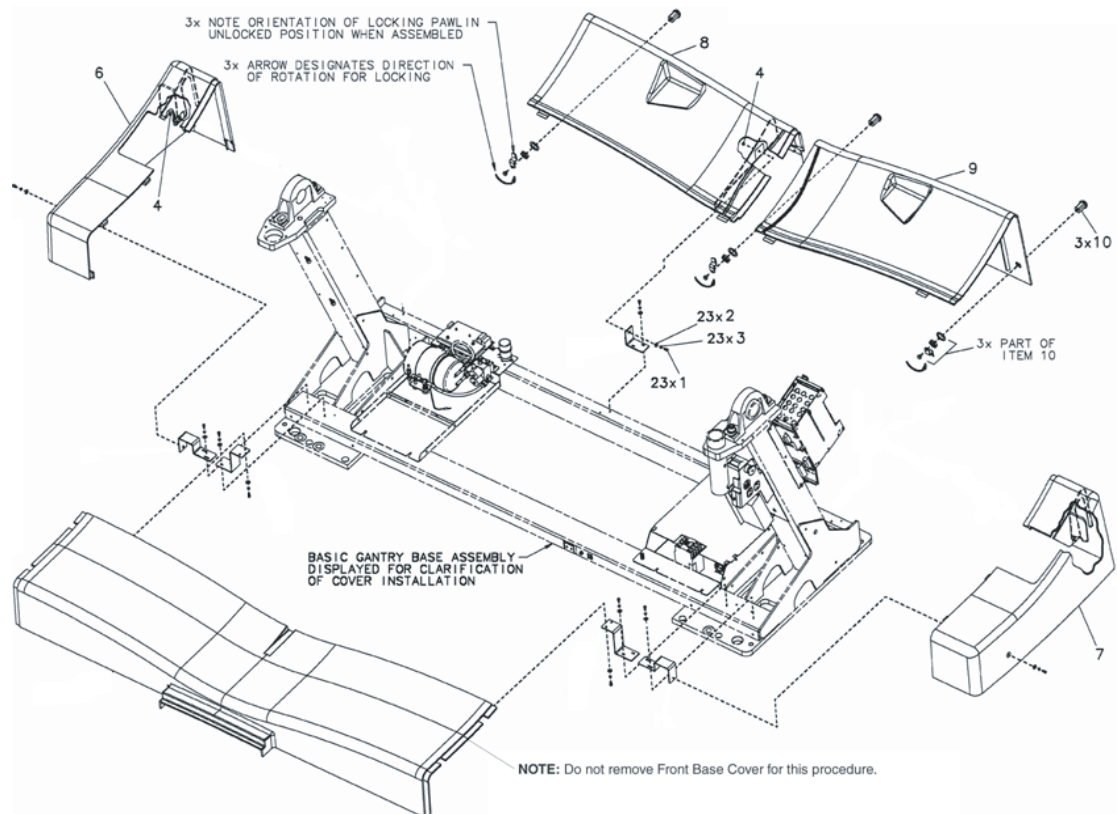


Figure 1-1 Gantry Base Covers

3.2 Install LAN Switch

- 1.) Attach VELCRO Tape to the LAN switch and the mounting bracket as shown in illustration below.

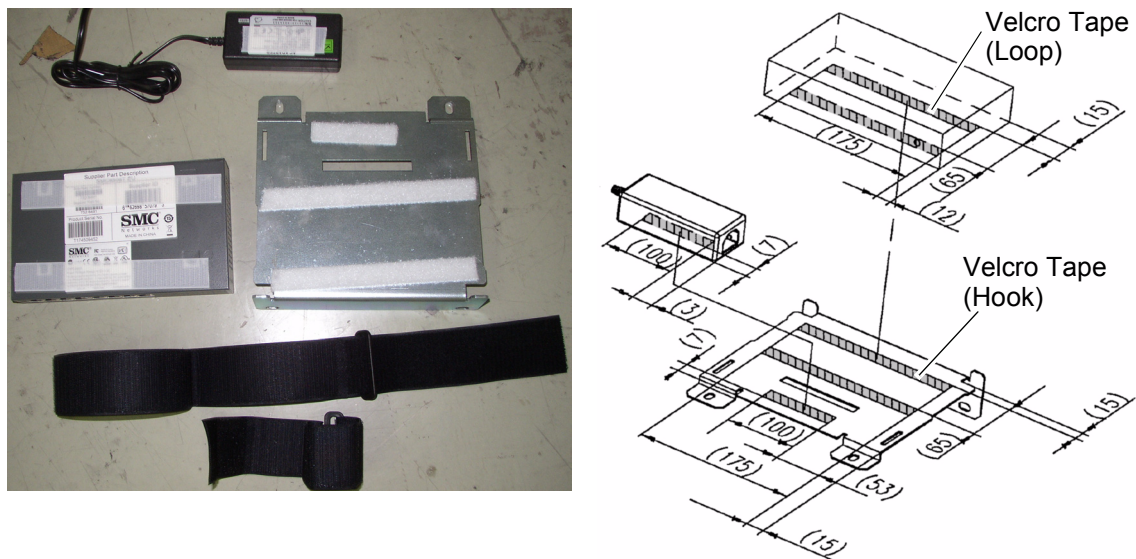


Figure 1-2 Velcro Tape Attachment

- 2.) Fix the LAN switch and the AC/DC adapter to the mounting bracket with long and short Velcro tapes.



Figure 1-3 LAN Switch Assembly

- 3.) Install the LAN switch assembly to the Gantry rear base.

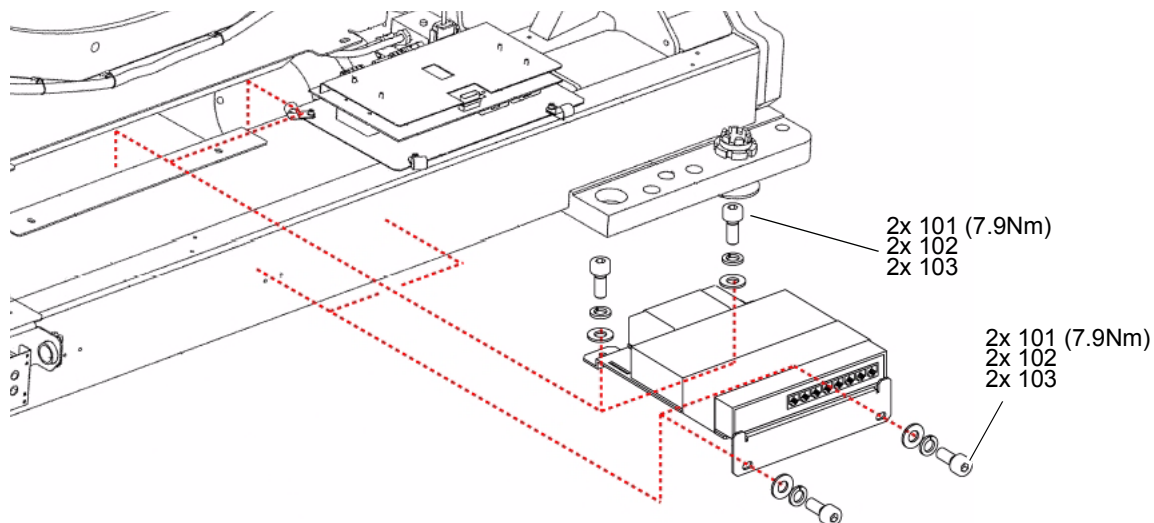


Figure 1-4 LAN Switch Assembly Installation

3.3 Install IPC and Accessory Panel

- 1.) Assemble the IPC and the left accessory panel.

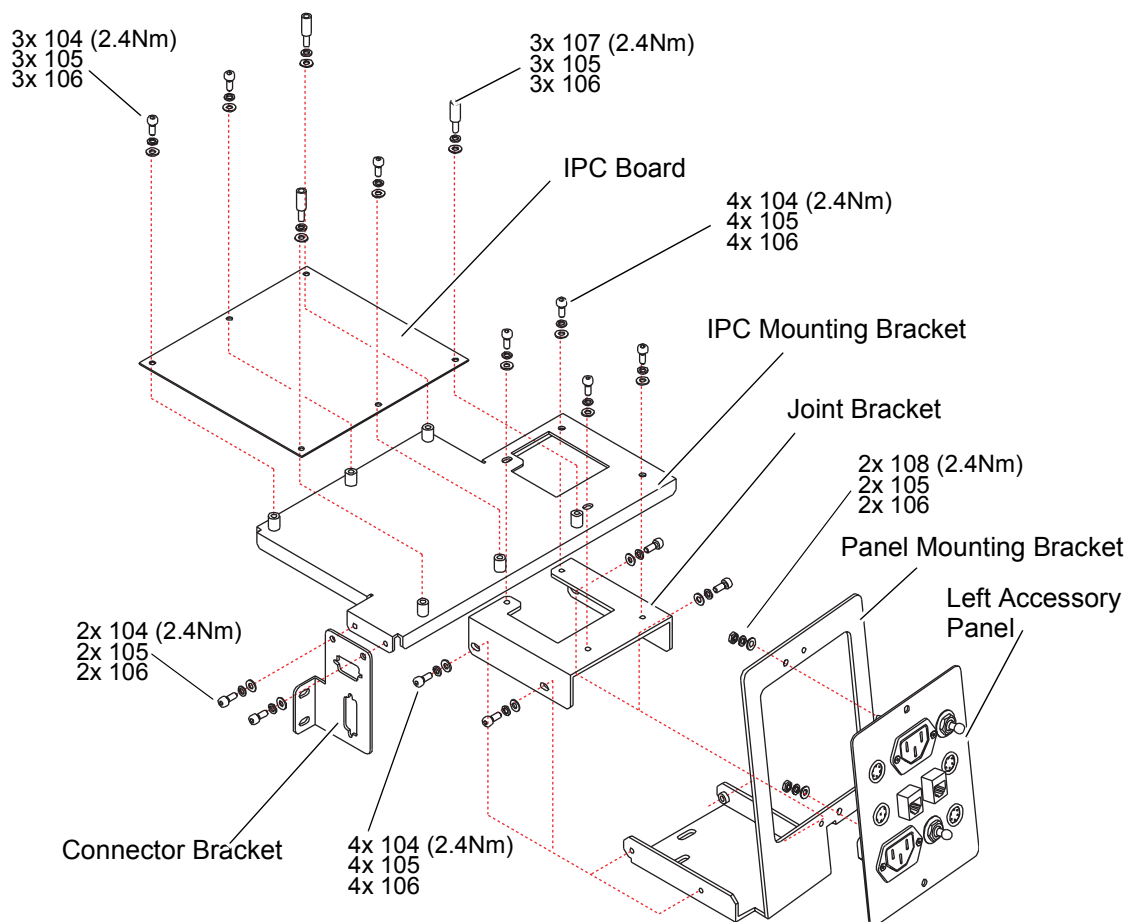


Figure 1-5 IPC Board and Left Accessory Panel

- 2.) Install the IPC assembly and left panel to the Gantry rear base.

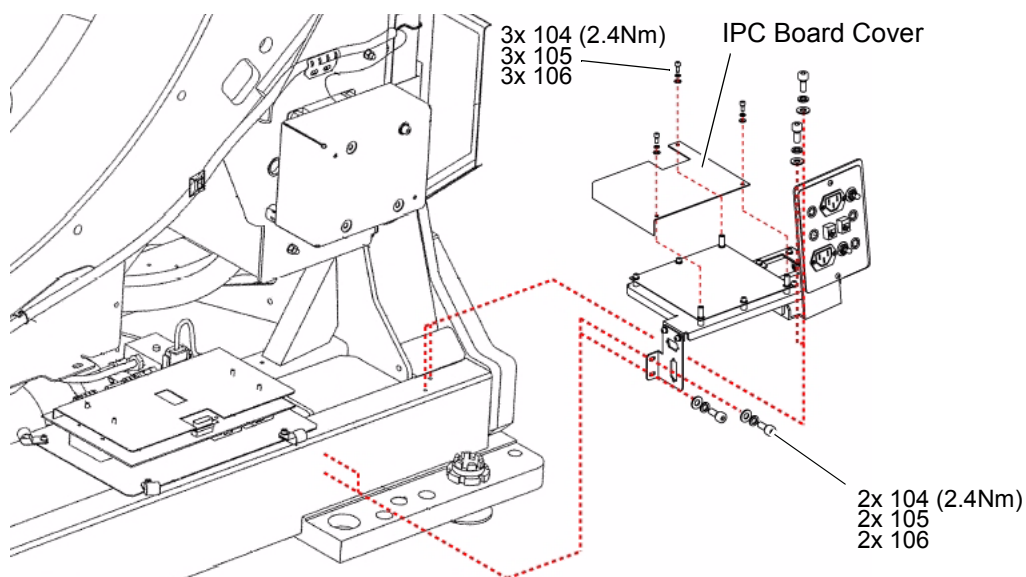


Figure 1-6 IPC and Left Accessory Panel Installation

- 3.) Connect 5306201 cable (Connector Bracket <-> IPC, J11 <-> Left Accessory Panel).

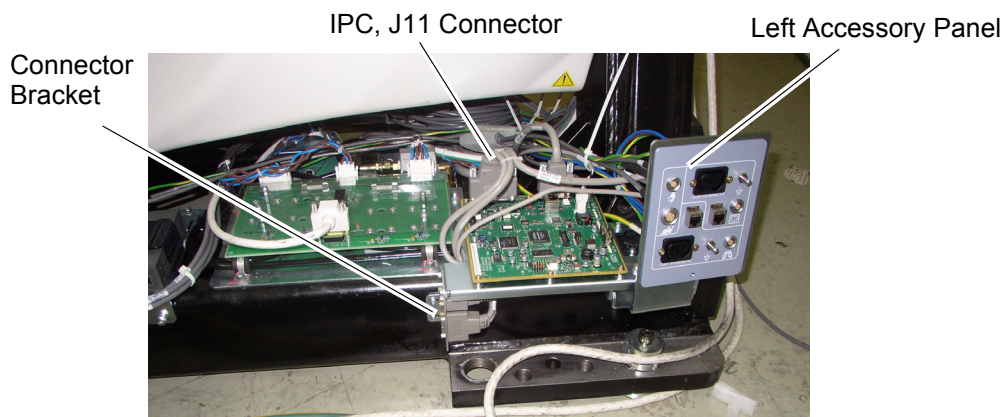
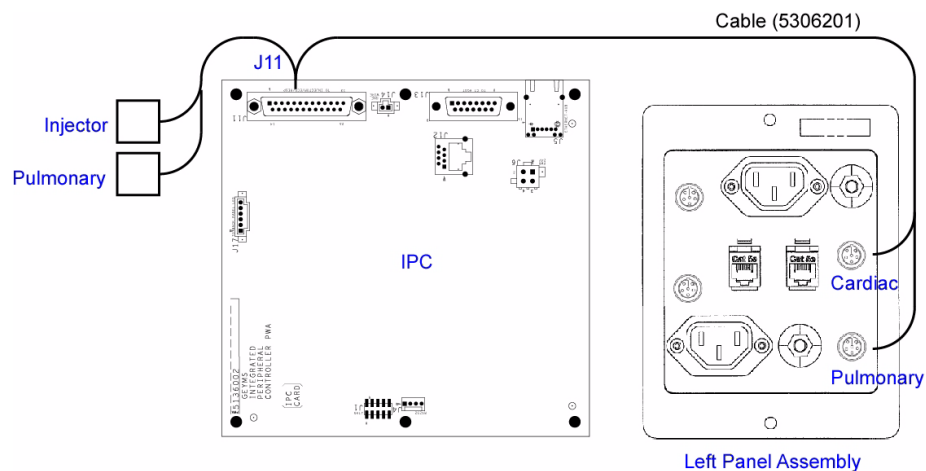


Figure 1-7 Cable (5306210) Connection

4.) Assemble the right accessory panel, and install it to the Gantry rear base.

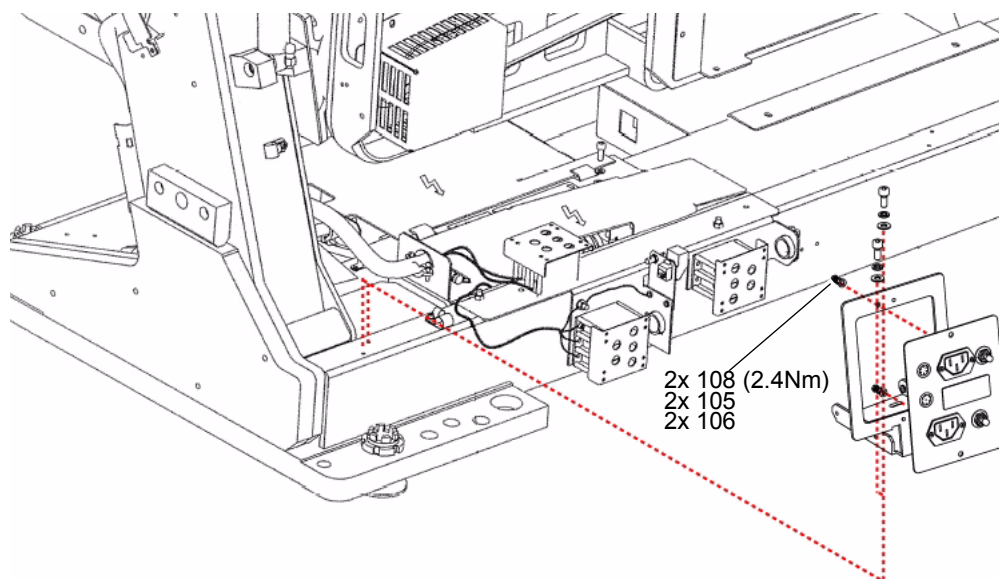


Figure 1-8 Right Accessory Panel Installation

3.4 Install Cable Harness

- 1.) Carefully route the cable harness as shown in illustration [Figure 1-9](#) and [Figure 1-10](#).

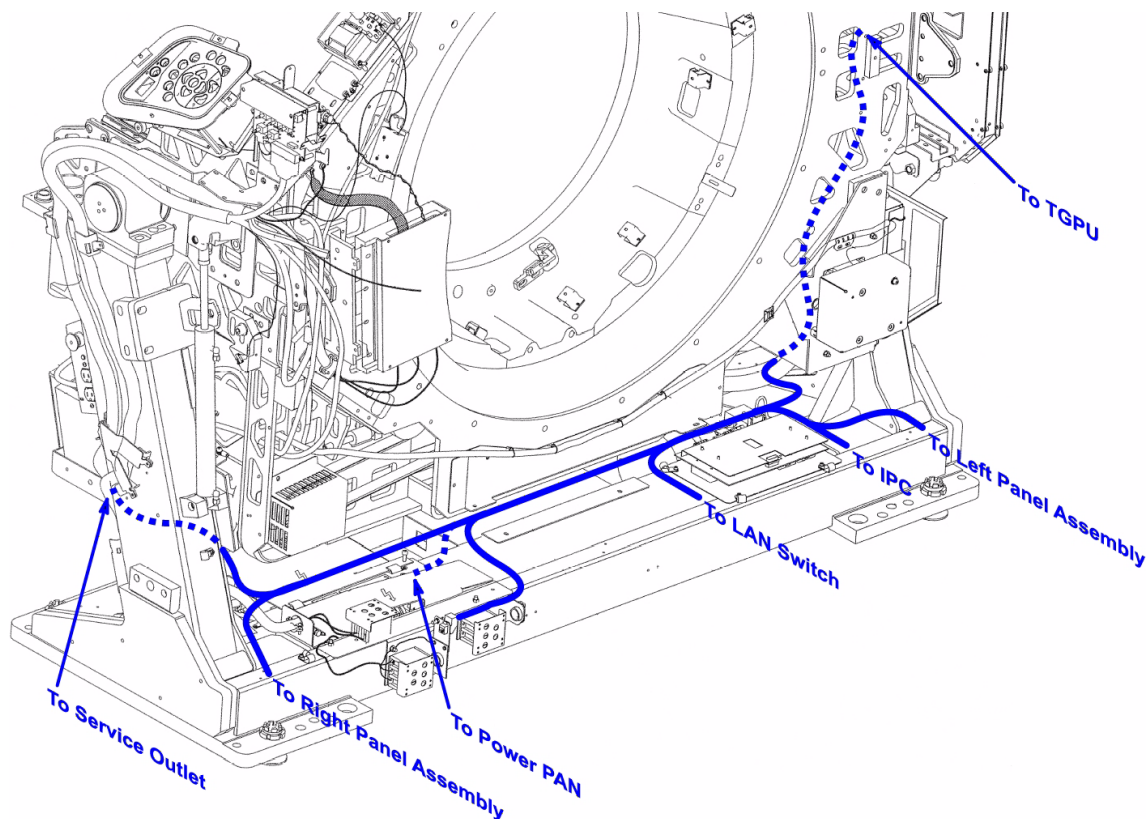


Figure 1-9 Cable Wiring

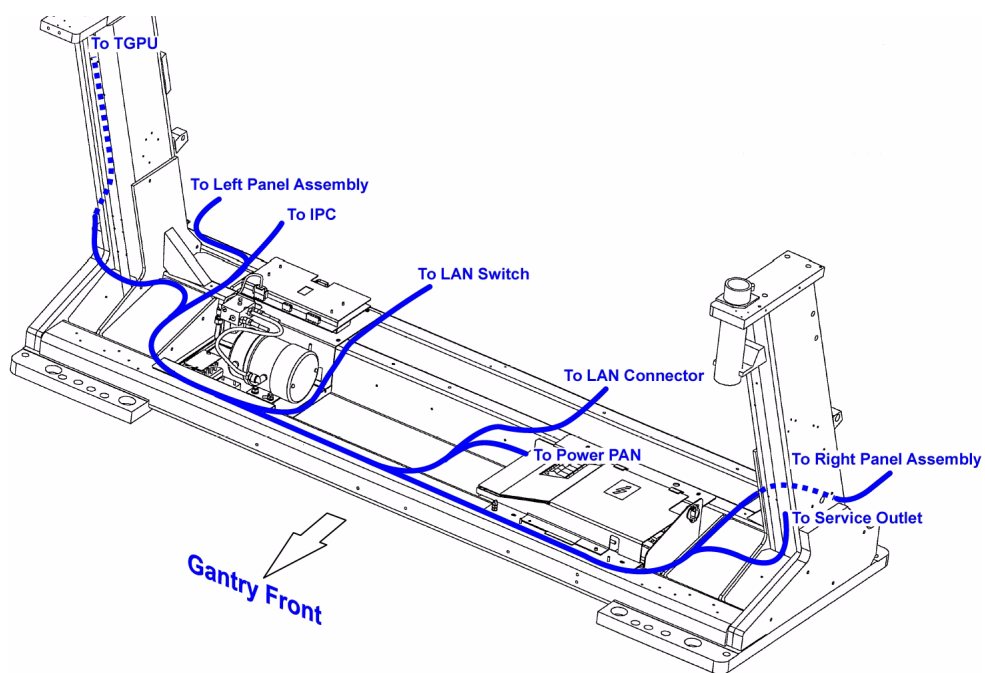


Figure 1-10 Cable Routing

- 2.) Connect all cable connectors of cable harness as shown in [Figure 1-11](#).
- (Refer to [Figure 1-12](#) through [Figure 1-16](#) for further information)

Cable Harness Assembly (5316429) Connection

Item	Part No.	Description	Cable Connection
2	5189456	Cable Asm - IPC power	IPC, J6 <=> TGP, J1 <=> Power Connector
4	5169215	cable, cardiac-pulmonary_PathFinder	IPC, J13 <=> TGPU, J22
5	2286150-2	Smart step cable for BSD Delight Lite.	R-Panel, J2, J3 <=> TGPU, J27 <=> L-Panel, J2, J3
7	5265804-2	Cable Asm-Network Switch Power, BSD V3	Service Outlet <=> Harness Outlet <=> LAN Switch
8	5306481	Power Cable, IPC Accessory Panel, VCT HD	R-Panel <=> Power Pan <=> L-Panel
10	5193969-9	Cable-LAN, BSD Delight Lite	Power Pan <=> LAN Switch 3
12	5193969-8	Cable-LAN, BSD Delight Lite	IPC, J5 <=> LAN Switch 2
13	5193967-2	TGPU J5 to LAN Switch Port1, BSD Delight Lite	TGPU, J5 <=> LAN Switch 1

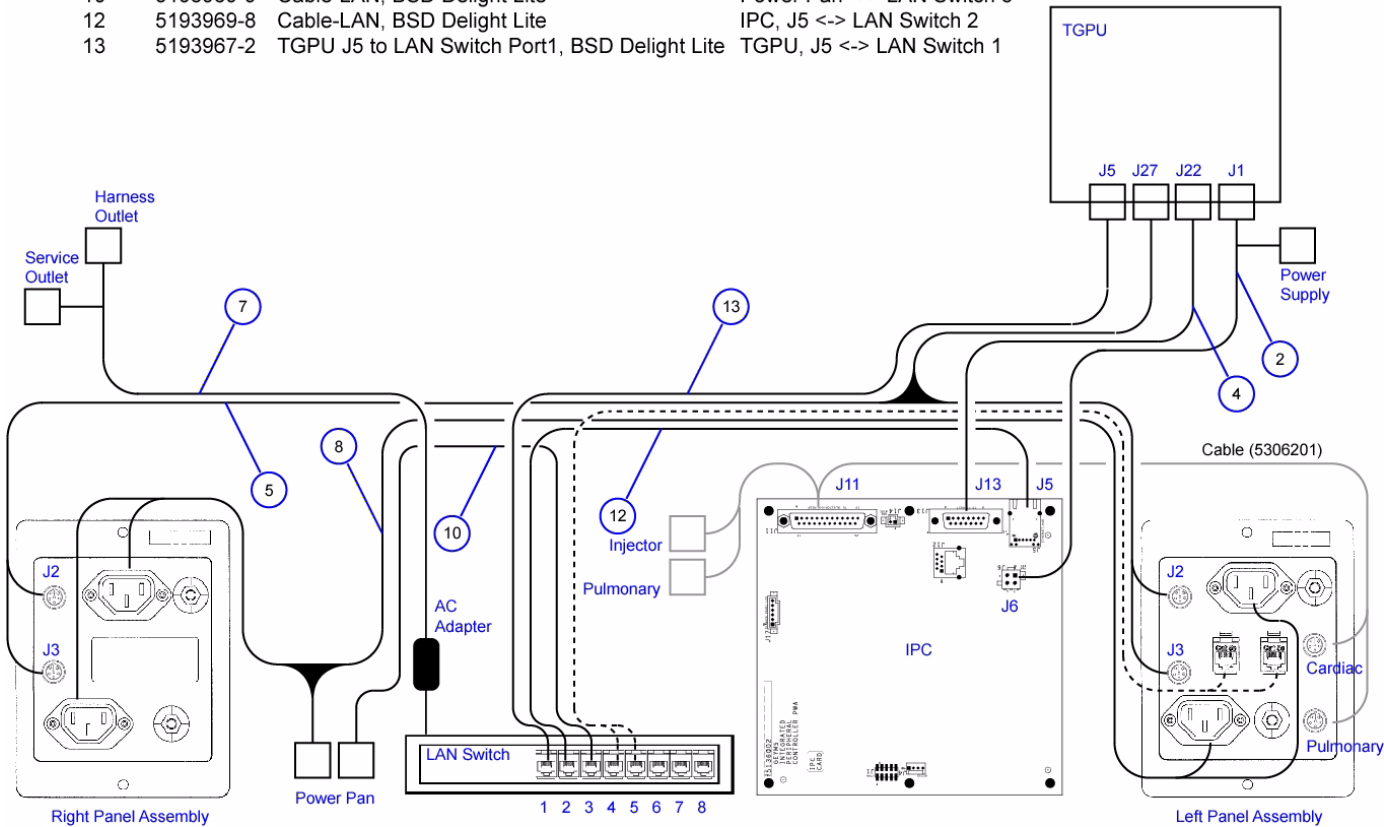


Figure 1-11 Cable Connections

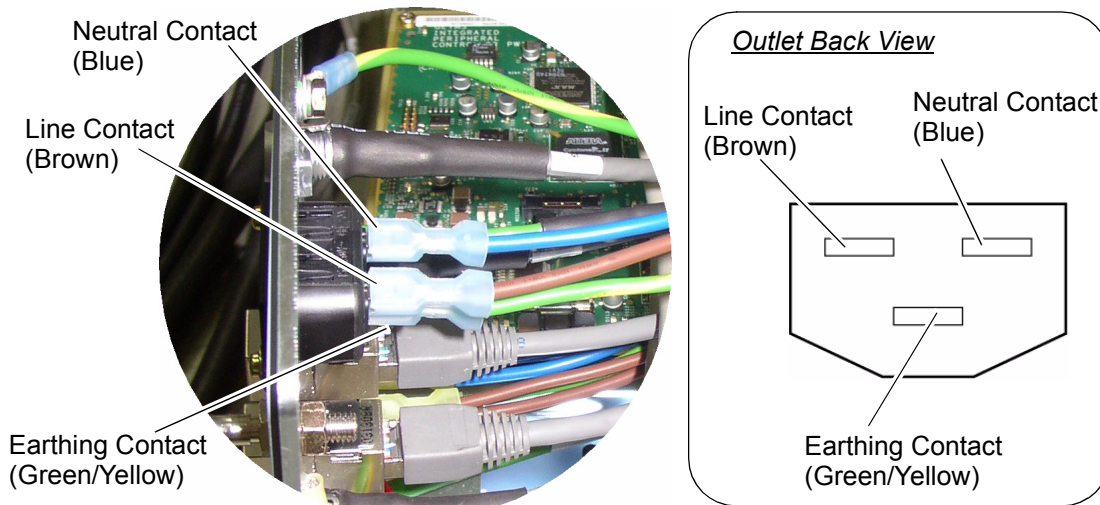
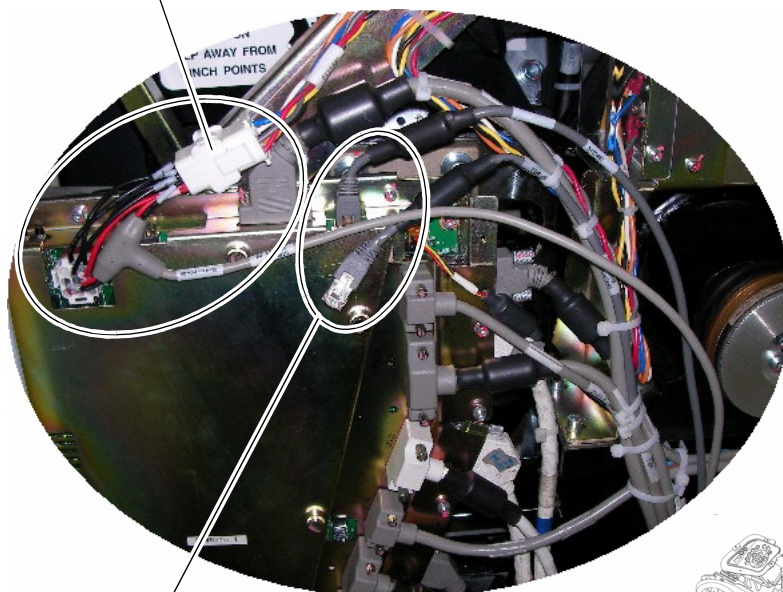


Figure 1-12 Outlet Connection

Disconnect the power cable connector from TGPU, then connect *Item2* cable connector to it.



Disconnect the LAN cable connector from TGPU, then connect the *Item13* cable connector to TGPU.

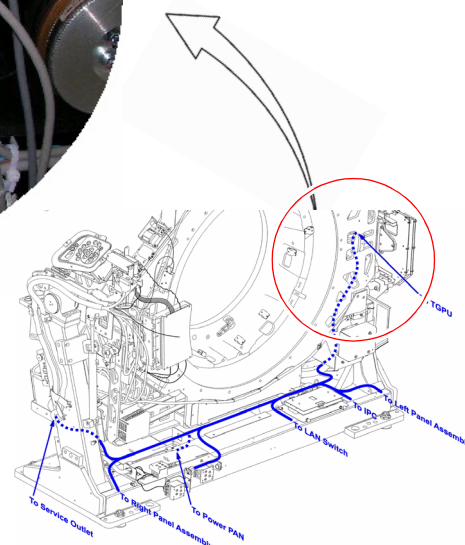


Figure 1-13 TGPU Cable Connection

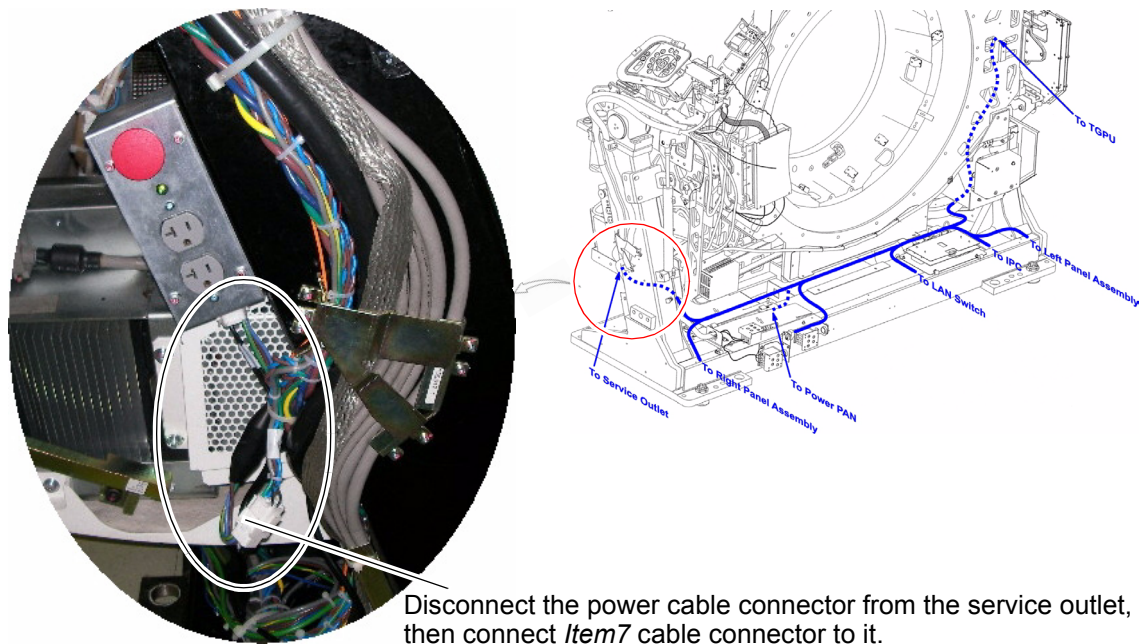


Figure 1-14 Service Outlet Cable Connection

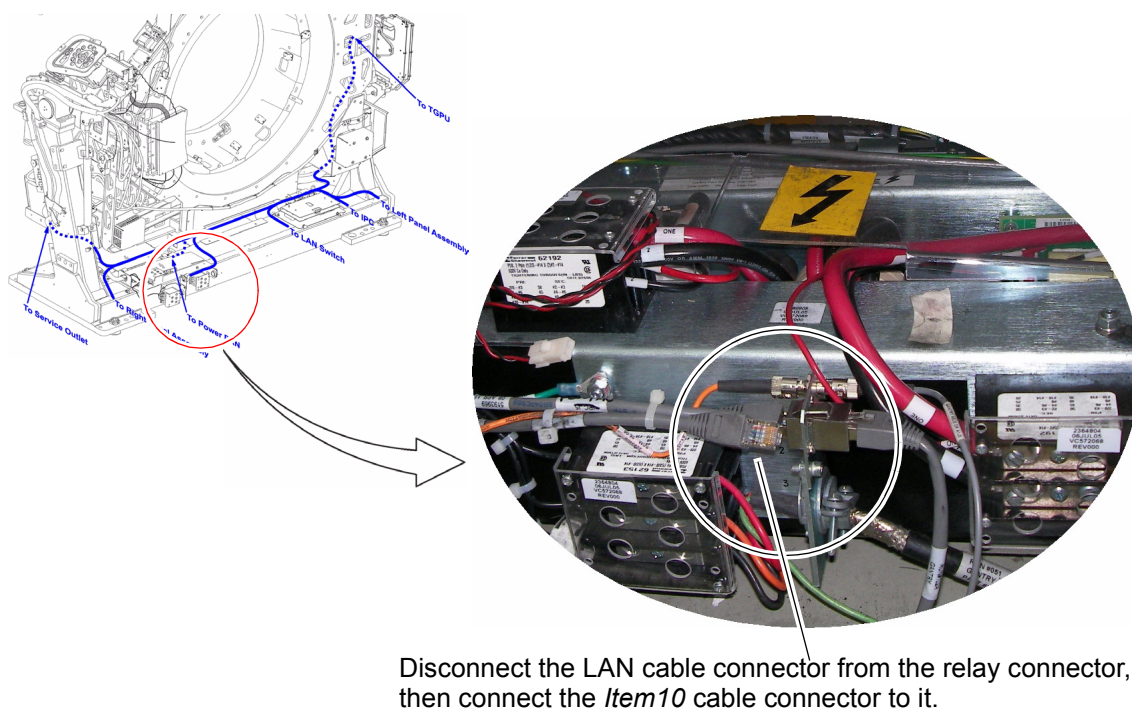


Figure 1-15 Power PAN LAN Cable Connection

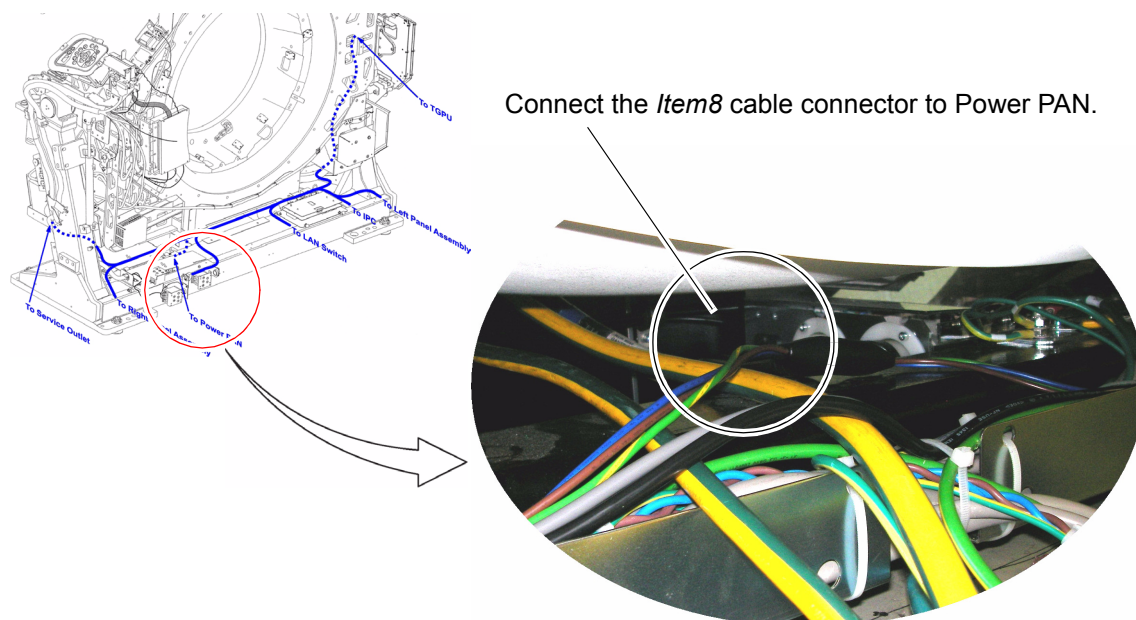


Figure 1-16 Power PAN Cable Connection

- 3.) Secure the cables with tie wraps.
- 4.) Restore power to the system.

Section 4.0 Finish Installation

4.1 LED Inspection of IPC Board

- 1.) To make sure the IPC board is functioning, an LED inspection should be made on the board.
The LED configuration on the board should be as follows with power on but no options connected to the options interface panels:

LED	Color	Label	Status (without options connected)
DS1	Green	24V_PWR	On Solid
DS2	Green	PHY_FW_LED	Blinking with network activity
DS3	Green	HRTBT	Blinking
DS4	Green	FW_LED_0	Off
DS5	Green	FW_LED_1	Off
DS6	Green	FW_ERROR	Off
DS7	Green	24V_FUSE	On Solid
DS8	Green	RESET	On Solid
DS9	Yellow	RESET	Off
DS10	Green	R_PULSE_FILTERED	Off

Table 1-2 IPC Board LED Status

LED	Color	Label	Status (without options connected)
DS11	Green	RESP_PULSE_FILTERED	Off
DS12	Green	R_PULSE_UNFILTERED	Off
DS13	Green	RESP_PULSE_UNFILTERED	Off
DS14	Yellow	FPGA_LOAD	Off
DS15	Green	FPGA_LOAD	On Solid
DS16	Yellow	B/A_CODE	Off
DS17	Green	B/A_CODE	On Solid
DS18	Green	+1.2V_OK	On Solid
Ethernet	Orange/Green	Ethernet-Hub	Blinking with network activity

Table 1-2 IPC Board LED Status

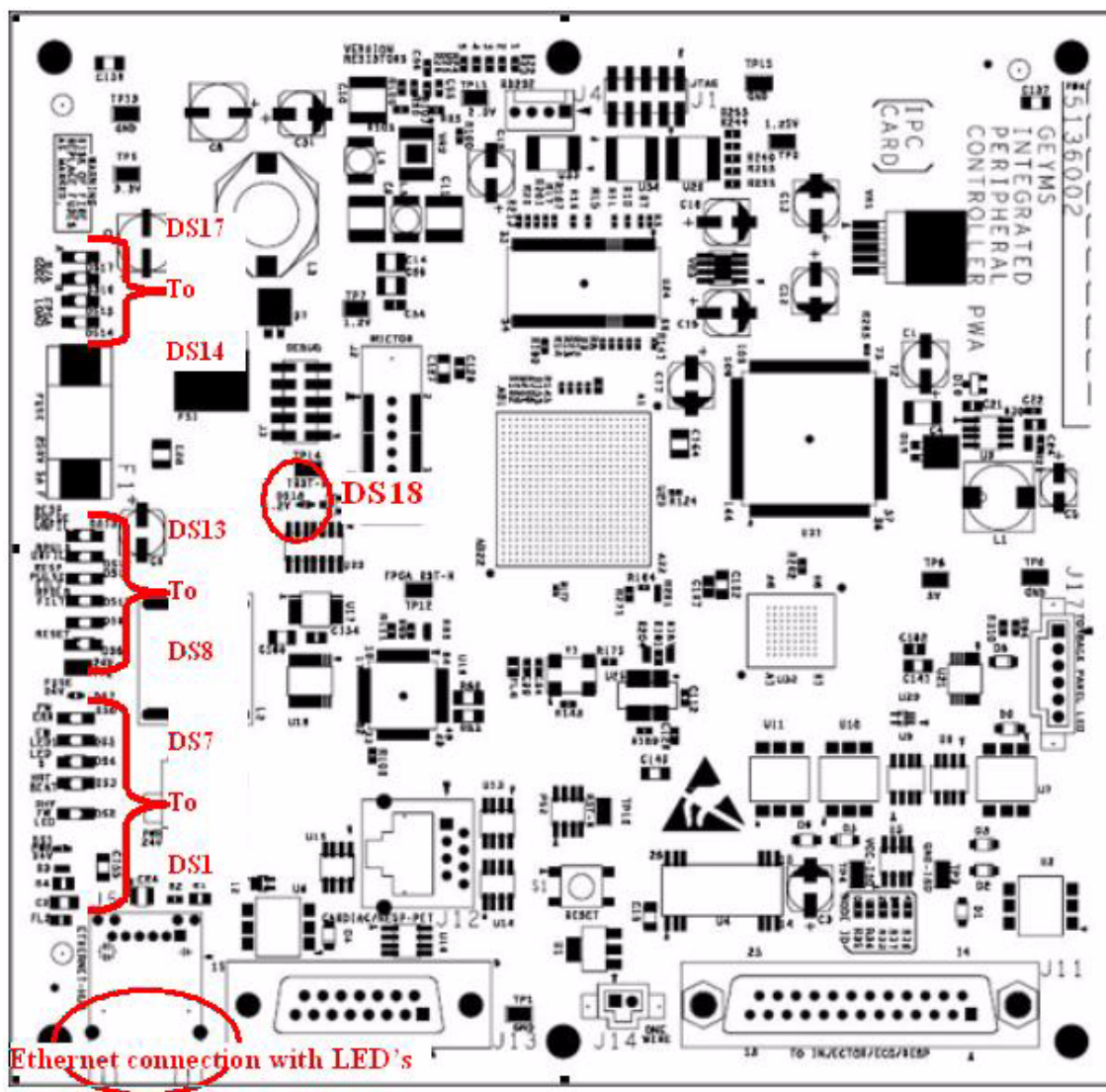


Figure 1-17 IPC Board LED locations

4.2 LED Inspection of LAN Switch

- 1.) Visually verify the front panel Power LED of the switch is illuminated.
- 2.) Visually verify that all appropriate Port LEDs show active links.

4.3 Finalization

- 1.) Perform an application scan to ensure proper system operation before turning the system over to the customer.
- 2.) Turn OFF all 3 switches (Axial Drive, HVDC, 120 VAC) on the service switch panel.
- 3.) Reinstall both rear base covers (see [Figure 1-1](#)).
- 4.) Reinstall the gantry side covers. Refer to your System Service Manual for any needed instruction.

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Imagination at work